

ADVANCE

SIMULATED ADVANCE — brownfield specification upgrade

The simulated Ministry-grade evidence resolves a sustained target-specification gap, establishes incremental adjacency, and passes minimum-support, national-value and competition controls.

SAU-H0-721049 · Conditional brownfield debottlenecking / line expansion · Mode: Ministry simulation

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Non-corrugated zinc-coated flat-rolled iron or non-alloy steel, width 600 mm or more
HS H0 / 721049

Unchanged public conclusion
INVESTIGATE — binding constraint unresolved

Target-specification demand 104 kt	Effective qualified capacity 57.509 kt
Published capability distance D* 0.2667 ratio	Minimum effective support S* 18 SAR m

Decision conditions
Achieve customer acceptance for the named target specification. [2 more; see the full record](#)

Kill conditions
Committed target-specification demand falls below 72 kt. [2 more; see the full record](#)

Evidence passport ledger: [S-WITS-721049](#) [S-UNICOIL-EPD](#)

Public evidence contradictions: [S-UNICOIL-SPEC](#)

Snapshot: PUBLIC-SAU-H0-721049-2026-08-31 · As of: 2026-08-31

Supporting evidence pack

1. [Identity and specification boundary](#)
2. [Demand and public trade signals](#)
3. [Domestic supply and qualified capacity](#)
4. [Gap diagnosis and false-positive controls](#)
5. [Capability and route alternatives](#)
6. [Economics and intervention](#)
7. [Competition and public policy](#)
8. [Complete rule execution ledger](#)
9. [Evidence passports and contradictions](#)
10. [Conditions, kill conditions and next facts](#)
11. [Authority and reproducibility](#)

Identity and specification boundary

Class D simulation · SYN-MINISTRY-STEEL-001

SIMULATED — NOT MINISTRY EVIDENCE

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

Unchanged public conclusion

INVESTIGATE — binding constraint unresolved

Brownfield priority to test

Test brownfield first; greenfield is not justified from public evidence.

Decision conditions

- No greenfield or financial support recommendation before effective capacity and target specification are resolved.

Kill conditions

- Demand is temporary, re-exported or below efficient scale
- Incumbent expansion already closes the specification-adjusted gap

Next evidence actions

- Line-level production by specification, application and destination
- Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages
- Importer/offtaker specifications and reason local supply was not selected
- Re-export and domestic-origin flow decomposition

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

- Delivered cost, import parity and unsupported Brownfield economics

Product identity

Source field	Value	Unit
HS revision	H0	
HS code	721049	
National tariff line	Unavailable Not established by the supplied evidence.	
Commercial name, English source	Non-corrugated zinc-coated flat-rolled iron or non-alloy steel, width 600 mm or more	
Commercial name, Arabic	منتجات مسطحة مدرفلة من حديد أو صلب غير مخلوط، مطلية بالزنك، بعرض 600 مم أو أكثر، غير مموجة	
Application boundary	Multiple applications; target customer/application remains unresolved in public evidence.	
Target specification	Z275 structural galvanized coil for coastal construction SYN-MINISTRY-STEEL-001::target_specification	
Standard	SASO-ASTM A653/A653M SYN-MINISTRY-STEEL-001::target_specification	
Alloy	Unavailable Not established by the supplied evidence. SYN-MINISTRY-STEEL-001::target_specification	
Temper	Unavailable Not established by the supplied evidence. SYN-MINISTRY-STEEL-001::target_specification	
Application	coastal construction panels SYN-MINISTRY-STEEL-001::target_specification	
Zinc coating mass	275 SYN-MINISTRY-STEEL-001::target_specification	g/m ²
Tin coating mass	Unavailable Not established by the supplied evidence. SYN-MINISTRY-STEEL-001::target_specification	g/m ²
Thickness	0.7 1.5 SYN-MINISTRY-STEEL-001::target_specification	mm

Source field	Value	Unit
Thickness	Unavailable Not established by the supplied evidence. SYN-MINISTRY-STEEL-001::target_specification	µm
Width	• 1000• 1250 SYN-MINISTRY-STEEL-001::target_specification	mm
Customer qualification required	Yes SYN-MINISTRY-STEEL-001::target_specification	

Classification revision and concordance

Unavailable

Specification source spans

Unavailable

Demand and public trade signals

Class D simulation · SYN-MINISTRY-STEEL-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Demand conclusion

Gross imports are a screening signal, not demand at the target specification.

Source field	Value	Unit
Base demand	Unavailable Not established by the supplied evidence. SYN-MINISTRY-STEEL-001::demand	kt
Target-specification demand	104 SYN-MINISTRY-STEEL-001::demand	kt
Downside demand	100 SYN-MINISTRY-STEEL-001::demand	kt
Committed demand	74 SYN-MINISTRY-STEEL-001::demand	kt

Source field	Value	Unit
Announced demand	22 SYN-MINISTRY-STEEL-001::demand	kt
Commitment probability	Unavailable Not established by the supplied evidence. SYN-MINISTRY-STEEL-001::demand	ratio
Observation year	2024	year
Gross import value	236.9	USD m
Gross import quantity	287.9	kt
Named buyers	Unavailable Not established by the supplied evidence.	
Buyer concentration	Unavailable Not established by the supplied evidence.	
Demand timing	Unavailable Not established by the supplied evidence.	
Year	Import value, USD m	Import quantity, kt
2021	194.1	177.4
2023	186	173.8
2024	236.9	287.9

Trade coverage and measurement

Trade-flow basis	gross
Re-exports separated	No
Domestic-origin exports separated	No
Missing years	• 2022
Monthly partner/tariff-line detail available	No
Quantity is comparable	Yes
Execution ceiling	DEGRADED where continuity or tariff-line detail is required

Production, retained flows and re-exports

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Reference year	2024
Domestic production (kt)	Unavailable
Retained imports (kt)	Unavailable
Domestic-origin exports (kt)	Unavailable
Re-exports (kt)	Unavailable
Source evidence references	None recorded; no completed review is implied.

Supplier diagnostics

Partner concentration by value (HHI)	0.36
Largest supplier share	Not calculable
Top-two supplier value share	0.763
Partner concentration by quantity (HHI)	Not calculable
Largest supplier quantity share	Not calculable
Bulk unit-value range (USD/t)	776 864

Small-volume high-unit-value observation

Trading partner	Austria
Quantity (kt)	1
Unit value (USD/t)	4179

Domestic supply and qualified capacity

Supply conclusion

Public producer observations; nameplate is not qualified capacity.

Named producer	Value
UNICOIL	Process family coated_steel
	Published process route continuous hot-dip galvanising
	Published standards SASO-ASTM A653/A653M
	Published coating range (g/m²) 45 350
	Observed nameplate capacity, t/y 250000
	Nominal-capacity status observed
	Nominal-capacity evidence reference S-UNICOIL-EPD
	Evidence class C
	S-UNICOIL-EPD S-UNICOIL-SPEC

Named producer	Value
Hadeed	Process family coated_steel
	Published process route cold-rolled galvanised and colour-coated coil publicly listed
	Published standards Unavailable
	Published coating range (g/m²) Unavailable
	Observed nameplate capacity, t/y Unavailable
	Nominal-capacity status unresolved
	Nominal-capacity evidence reference Unavailable
	Evidence class C
S-HADEED	

Observed adjacency signals

• Signal type	core_process
Description	continuous hot-dip galvanising
Evidence references	◦ S-UNICOIL-EPD
• Signal type	core_process
Description	cold rolling and pickling
Evidence references	◦ S-UNICOIL-EPD
• Signal type	adjacent_output
Description	published ASTM/SASO grades
Evidence references	◦ S-UNICOIL-EPD◦ S-UNICOIL-SPEC

• Signal type	relevant_certification
Description	ISO/IEC 17025 laboratory
Evidence references	◦ S-UNICOIL-EPD

Class D simulation · SYN-MINISTRY-STEEL-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value	Unit
Simulated nameplate capacity	250 SYN-MINISTRY-STEEL-001::plant_line	kt
Availability factor	0.92 SYN-MINISTRY-STEEL-001::plant_line	ratio
Yield factor	0.94 SYN-MINISTRY-STEEL-001::plant_line	ratio
Qualification share	0.38 SYN-MINISTRY-STEEL-001::plant_line	ratio
Market allocation share	0.7 SYN-MINISTRY-STEEL-001::plant_line	ratio
Current utilisation	0.89 SYN-MINISTRY-STEEL-001::plant_line	ratio
Description	Brownfield finishing-control, coating-mass stability and customer-qualification package SYN-MINISTRY-STEEL-001::upgrade	
Incremental capacity	50 SYN-MINISTRY-STEEL-001::upgrade	kt
Upgrade schedule	18 SYN-MINISTRY-STEEL-001::upgrade	months
Effective qualified capacity	57.509	kt
Qualified available supply	57.5092	kt

Gap diagnosis and false-positive controls

Class D simulation · SYN-MINISTRY-STEEL-001

SIMULATED — NOT MINISTRY EVIDENCE

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

Quantity gap

Source field	Value	Unit
Specification-adjusted gap	46.491	kt

Source reference	Status	Value
EX-01_HETEROGENEOUS_RESIDUAL	Not satisfied	The commercial product cannot be separated from a highly heterogeneous residual code and no product-level evidence is available. SYN-MINISTRY-STEEL-001::hard exclusion inputs
EX-02_MARKET_BELOW_MES	Not satisfied	The sustainable downside market is below minimum efficient scale and no credible export contract exists. SYN-MINISTRY-STEEL-001::hard exclusion inputs
EX-03_UNSATISFIABLE_HARD_GATE	Not satisfied	A legal, safety, environmental, IP or customer-qualification hard gate cannot be satisfied. SYN-MINISTRY-STEEL-001::hard exclusion inputs
EX-04_IDLE_EQUIVALENT_CAPACITY	Not satisfied	Domestic production meets the required specification with material idle capacity and no binding market failure is evidenced. SYN-MINISTRY-STEEL-001::hard exclusion inputs

Source reference	Status	Value
EX-05_TRANSITORY_OR_MEASUREMENT	Not satisfied	The apparent gap is dominated by re-export, one-off project demand, temporary price arbitrage or classification discontinuity. SYN-MINISTRY-STEEL-001::hard_exclusion_inputs
EX-06_REDUNDANCY_OR_CROWD_OUT	Not satisfied	The intervention would create unacceptable redundant capacity or crowd out a more efficient incumbent. SYN-MINISTRY-STEEL-001::hard_exclusion_inputs

Test the apparent gap

Source reference	Status	Value
hard_exclusion	Not satisfied	HARD_EXCLUSION_NOT_SATISFIED SYN-MINISTRY-STEEL-001::hard_exclusion_inputs
false_or_measurement_gap	Not satisfied	FALSE_OR_MEASUREMENT_NOT_ESTABLISHED SYN-MINISTRY-STEEL-001::hard_exclusion_inputs
equivalent_idle_qualified_supply	Not satisfied	EQUIVALENT_QUALIFIED_SUPPLY_NOT_ESTABLISHED SYN-MINISTRY-STEEL-001::demand SYN-MINISTRY-STEEL-001::equivalence
uneconomic_at_efficient_scale	Not satisfied	ECONOMIC_ROUTE_EXISTS SYN-MINISTRY-STEEL-001::route_evidence
structural_overcapacity	Not satisfied	STRUCTURAL_OVERCAPACITY_NOT_ESTABLISHED SYN-MINISTRY-STEEL-001::hard_exclusion_inputs

Source reference	Status	Value
generic_capacity_contradicted	Not satisfied	GENERIC_CAPACITY_NOT_CONTRADICTED S-UNICOIL-EPD

Capability and route alternatives

Class D simulation · SYN-MINISTRY-STEEL-001

SIMULATED — NOT MINISTRY EVIDENCE

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

Source field	Value	Unit
Known coverage K	1	ratio
Unknown weight U	0	ratio
Uncertainty penalty (lambda)	0.5	ratio
Minimum known coverage	0.7	ratio
Published capability distance D*	0.2667	ratio

Source field	State	Weight
Feedstock chemistry	0	0.05
Core process route	0	0.2
Equipment operating envelope	1	0.2
Finishing and specification control	2	0.15
Quality, laboratories and metrology	0	0.1
Certification and customer qualification	2	0.1
Capacity and timing	1	0.1

Source field	State	Weight
Utilities, environment and permitting	0	0.05
Skills and market integration	0	0.05

Sector-profile hard gates

Substrate range	RESOLVED
Width/thickness envelope	RESOLVED
Coating process and mass	RESOLVED
Surface treatment	RESOLVED
Mandatory or customer standard	RESOLVED

Unresolved hard gates • None recorded; no completed review is implied.

Known hard-gate failures • None recorded; no completed review is implied.

No intervention

Fails

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Fails
Additionality	Passes
Policy permissibility	Passes

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

passes

- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

Administrative or regulatory correction

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable
Lower-route precedence: No lower route recorded	
Incremental national value	Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

Information and market linkage

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
-

Certification, testing and quality

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

Demand aggregation and offtake

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

Brownfield upgrade or expansion

Passes

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Passes
Additionality	Passes
Policy permissibility	Passes

Unsupported NPV -18

Unsupported IRR 0.095

Minimum effective support S* 18

Lower-route precedence: No lower route recorded

Incremental national value	198
Capacity ratio	1.0751
Warning threshold	1.25
Passes the recorded test	Yes

[SYN-MINISTRY-STEEL-001::route_evidence](#)

Technology partnership or joint venture

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: Brownfield upgrade or expansion

Incremental national value	Not calculable
----------------------------	----------------

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

Greenfield capacity

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: Brownfield upgrade or expansion

Incremental national value Not calculable

Not calculable

- Greenfield requires failed lower routes, D* above the configured band, demand at MES, and passing competition gates.
- A lower-intervention route fully resolves the binding constraint.

Shared enabling infrastructure

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- GRAPH_REQUIRED: route 8 requires the governed dependency graph and a positive UnlockValue.

Public route comparison

No intervention

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

Administrative or regulatory correction

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- Administrative, classification or regulatory barrier evidence is required.

Information and market linkage

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified information, linkage or matching constraint is required.

Certification, testing and quality

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified specification, certification, testing or qualification constraint is required.

Demand aggregation and offtake

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable
Lower-route precedence: No lower route recorded	
Incremental national value	Not calculable
Not calculable	
A verified demand-fragmentation or offtake constraint is required.	

Brownfield upgrade or expansion

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded	
Incremental national value	Not calculable
Not calculable	
- Verified incumbent adjacency makes brownfield the priority hypothesis; effective capacity and economics remain required. - Downside route economics are unavailable.	

Technology partnership or joint venture

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified technology constraint and major-line/JV adjacency are required.

Greenfield capacity

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value	Not calculable
----------------------------	----------------

Not calculable

- Greenfield requires failed lower routes, D* above the configured band, demand at MES, and passing competition gates.

Shared enabling infrastructure

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV	Not calculable
-----------------	----------------

Unsupported IRR	Not calculable
-----------------	----------------

Minimum effective support S*	Not calculable
------------------------------	----------------

Lower-route precedence: No lower route recorded

Incremental national value	Not calculable
----------------------------	----------------

Not calculable

- GRAPH_REQUIRED: route 8 requires the governed dependency graph and a positive UnlockValue.

Counterfactual

Can the incumbent meet the specification without capital investment?	No
--	----

Does downside demand support both incumbent and entrant?	No
--	----

Does entry displace efficient domestic production?	Yes
--	-----

Is a technology partnership superior to expansion or greenfield?	No
Missing capabilities	• equipment_envelope • finishing_spec_control • certification_customer_qualification • capacity_time_window
Brownfield versus greenfield comparison	
Incremental capacity	50
Upgrade schedule	18
Unsupported NPV	-18
Minimum effective support S*	18
Greenfield alternative	Unavailable

Economics and intervention

Class D simulation · SYN-MINISTRY-STEEL-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value	Unit
Unsupported NPV	-18	SAR m
Unsupported IRR	0.095	ratio
Minimum effective support S*	18	SAR m
Supported NPV	0	SAR m
Supported IRR	0.12	ratio
Hurdle rate	0.12	ratio
Separate downside result	Not calculable Cannot be calculated from the supplied evidence.	SAR m

National-value comparison

National benefits	251
-------------------	-----

National costs	53
Incremental national value	198
Positive result	Yes
Components	
Domestic value added	126
Exports	24
Resilience value	62
Knowledge and skills	18
Fiscal receipts	21
Government cost	18
Displacement	12
Resources and environment	8
Risk allowance	15

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Currency	SAR_m
Cash flows without support (SAR m), in source order	-179.800920423• 28.6362006614• 28.6362006614 28.6362006614• 28.6362006614• 28.6362006614 28.6362006614• 28.6362006614• 28.6362006614 28.6362006614• 28.6362006614
Hurdle rate	0.12
Support instrument	conditional capex and qualification package
National value	
Domestic value added	126
Exports	24
Resilience value	62
Knowledge and skills	18

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Fiscal receipts	21
Government cost	18
Displacement	12
Resources and environment	8
Risk allowance	15
Minimum efficient scale (kt)	50

[SYN-MINISTRY-STEEL-001::economics](#)

Cash-flow inputs in source order

• Route code	5
Recorded basis	Brownfield finishing-control, coating-mass stability and customer-qualification package; downside economics, national value and competition inherited from the economics, upgrade and demand blocks.

[SYN-MINISTRY-STEEL-001::route_evidence](#)

Competition and public policy

Class D simulation · SYN-MINISTRY-STEEL-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value	Unit
Capacity to downside demand ratio	1.0751	ratio
Displacement cost	12	SAR m
Sunset date	Unavailable Not established by the supplied evidence.	
Capacity to downside demand ratio	1.0751	
Warning threshold	1.25	

Warning is triggered	No
Triggers the default warning	Yes
Displacement cost	12
Non-additionality assessment	
Status	passes
Recorded basis	route_record_additionality
Concentration before and after	
Not calculable	

Calculation and recommendation do not authorize public support.

Complete rule execution ledger

R0 · Identity gate

Full · Fires

HS revision H0 and HS6 721049 are frozen in the snapshot; specification/application remains separately gated.

Continue to screening while preserving the unresolved decision object.

R1-F · Persistent retained exposure — full

Disabled · Not evaluable

Full continuity is unavailable and gross flows are not retained imports.

Do not claim the full persistence rule.

Missing years • 2022

R1-D · Persistence signal — degraded

Degraded · Fires

3 positive observed years within the configured window; no interpolation used.

Generate INVESTIGATE only; confidence capped at C.

Positive observed years	• 2021• 2023• 2024
Confidence ceiling	C

R2 · Quantity-led expansion

Full · Fires

Quantity-led expansion signal fires.

Separate structural volume growth from price movement.

First year	2023
Final year	2024
Observed period (years)	1
Change in log value	0.2419
Change in log quantity	0.5047
Change in log unit value	-0.2625
Quantity contribution share	0.6579
Quantity compound annual growth rate	0.6565

R3 · Supplier concentration

Full · Fires

External supply is concentrated on the value basis; quantity concentration is NOT_CALCULABLE.

Generate a resilience/diversification review, not an automatic localisation recommendation.

Concentration index (HHI)	0.36
Largest supplier share	Not calculable
Top-two supplier share	0.763
Concentration threshold (HHI)	0.25
Largest-supplier threshold	0.5
Value	
Recorded basis	value

Status	FULL
Observation year	2024
Trade-flow basis	gross
Source	disclosed_concentration
Concentration index (HHI)	0.36
Largest supplier share	Not calculable
Top-two supplier share	0.763
Valid observation coverage	Not calculable
Eligible partner count	Not calculable
Source explanation	Unavailable

Quantity	
Recorded basis	quantity
Status	Not calculable
Observation year	2024
Trade-flow basis	gross
Source	none
Concentration index (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier share	Not calculable
Valid observation coverage	Not calculable
Eligible partner count	Not calculable
Source explanation	No partner rows or source-attributed quantity concentration disclosure are available.

R4-F · Product-tier heterogeneity — full

Disabled · Not evaluable

Partner-month × tariff-line cells are unavailable.

Do not fit clusters or claim grades.

R4-D · Unit-value dispersion — degraded

Degraded · Fires

A source-attributed annual unit-value dispersion summary supports a descriptive product-mix signal; no cluster or grade conclusion.

Open specification research only; no grade conclusion.

Status	DISCLOSED
Calculation basis	disclosed_dispersion
Valid value coverage	Not calculable
Valid quantity coverage	Not calculable
Minimum valid-value coverage	0.7
Weighted first quartile (USD/t)	Not calculable
Weighted median (USD/t)	Not calculable
Weighted third quartile (USD/t)	Not calculable
Interquartile range (USD/t)	Not calculable
Bulk unit-value band (USD/t)	• 776 • 864
Potential outlier	
Trading partner	Austria
Net weight (kt)	1
Unit value (USD/t)	4179
Confirmed outlier	No
Comparison observations	Not calculable
Source evidence reference	S-WITS-721049
Source explanation	Annual partner data permit only the degraded R4-D path; complete row-level value and comparable-quantity coverage are not present in the frozen fixture.

R5 · Domestic supply plus continued imports

Degraded · Fires

Verified domestic capability coexists with material gross imports.

Test specification, qualification, capacity, price, application and allocation mismatch.

Retained imports (kt)	Not calculable
Net import exposure (kt)	Not calculable
Apparent consumption (kt)	Not calculable
Retained imports / apparent consumption	Not calculable
Calculation basis	Not calculable
Unavailable inputs	- domestic_origin_exports_kt • domestic_production_kt - reexports_kt • retained_imports_kt
Reasons for unavailability	
Retained imports (kt)	reexports_kt • retained_imports_kt
Net import exposure (kt)	domestic_origin_exports_kt • retained_imports_kt
Apparent consumption (kt)	- domestic_origin_exports_kt • domestic_production_kt - retained_imports_kt
Retained imports / apparent consumption	apparent_consumption_kt • retained_imports_kt
Configured threshold	0.2
Source explanation	Domestic production quantity and retained-import flow are absent from the frozen public snapshot; gross imports cannot establish apparent consumption.

R6 · Capacity pressure

Disabled · Not evaluable

Effective utilisation and target-specification shortage are not public.

Obtain line-level availability, yield, qualification share and allocation.

R7 · Latent domestic capacity

Disabled · Not evaluable

Idle qualified capacity and specification equivalence are not established publicly.

Do not assume latent capacity.

R8 · Committed future demand

Disabled · Not evaluable

Awarded/financed demand at the target specification is unavailable.

Keep base, committed and announced demand separate when provided.

R9-S · Coarse incumbent adjacency screen

Full · Fires

A verified Saudi plant is in the same process family with additional capability signals.

Open the full line-level capability assessment; do not publish D* from screening alone.

Same process family	Yes
---------------------	-----

Qualifying signal count	4
-------------------------	---

Known hard-gate failure	No
-------------------------	----

R10 · Strategic criticality

Degraded · Fires

A resilience review is warranted by concentration; formal criticality still requires authority confirmation.

Keep commercial viability and strategic value separate.

Criticality status	Not calculable
--------------------	----------------

Criticality evidence reference	Unavailable
--------------------------------	-------------

R11 · Economic exclusion / generic-capacity warning

Full · Does not fire

Gross exports are 0.1144× imports; the configured generic-capacity warning threshold is not met.

Reject generic support or move only a named specialty exception to investigation.

Export/import value ratio	0.1144
Export/import ratio threshold	50
Calculated export/import value ratio	0.1144
Disclosed export/import value ratio	Unavailable
Disclosed ratio is consistent	Unavailable
Ratio basis	gross_trade_value
Ratio status	CALCULABLE
Ratio explanation	Unavailable
Established nominal capacity	
Established by evidence	Yes
Total observed nominal capacity (t/year)	250000
Named producers	• UNICOIL
Evidence references	• S-UNICOIL-EPD
Source explanation	Unavailable

R12 · Evidence-value trigger

Degraded · Fires

5 named facts could change the route.

Prioritise the smallest evidence request with a plausible route effect; quantify EVSI when cost and value inputs exist.

Named missing facts	• Line-level production by specification, application and destination • Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages • Importer/offtaker specifications and reason local supply was not selected • Re-export and domestic-origin flow decomposition
---------------------	---

- Delivered cost, import parity and unsupported Brownfield economics

Evidence needed

- Line-level production by specification, application and destination

Evidence-need identifier	line-level production or producer-grade matrix
Blocked decision field	product_identity
Effect on route	Can establish domestic equivalence and change the incumbent route.
Evidence references	◦ S-UNICOIL-EPD◦ S-UNICOIL-SPEC
Numerical value of information	Not calculable

- Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages

Evidence-need identifier	capacity/availability/allocation
Blocked decision field	domestic_supply_or_capability
Effect on route	Can distinguish no action, brownfield expansion, and new capacity.
Evidence references	◦ S-UNICOIL-EPD
Numerical value of information	Not calculable

- Importer/offtaker specifications and reason local supply was not selected

Evidence-need identifier	target specification/application
Blocked decision field	demand_at_required_specification
Effect on route	Can establish target demand and change state or route.
Evidence references	◦ None recorded; no completed review is implied.
Numerical value of information	Not calculable

- Re-export and domestic-origin flow decomposition

Evidence-need identifier	re-export/origin decomposition
Blocked decision field	demand_at_required_specification

Effect on route	Can distinguish retained demand from a false or transitory gap.
Evidence references	◦ S-WITS-721049
Numerical value of information	Not calculable

- Delivered cost, import parity and unsupported Brownfield economics

Evidence-need identifier	route economics
Blocked decision field	route_economics
Effect on route	Can determine whether the plausible route proceeds unsupported.
Evidence references	◦ None recorded; no completed review is implied.
Numerical value of information	Not calculable

Simulated R6–R8 ledger

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

R5 · Retained import penetration

Disabled · Not evaluable

Synthetic retained-flow inputs are absent; R5 is not calculable.

Use only when production and retained-flow blocks are declared.

Retained imports (kt)	Not calculable
Domestic production (kt)	Not calculable
Gross import quantity	287.9
Retained-import penetration	Not calculable
Apparent consumption (kt)	Not calculable
Minimum retained-import penetration	0.2

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

R6 · Capacity pressure

Degraded · Fires

Synthetic utilisation and shortage proxy meet the configured R6 thresholds; sustained period is NOT_CALCULABLE.

Use as a DEGRADED capacity-pressure signal only; it cannot by itself support ADVANCE without a sustained period.

Effective utilisation	0.89
Minimum effective utilisation	0.85
Target-specification demand	104
Effective qualified capacity	57.509
Effective-capacity source field	effective_qualified_capacity_kt
Shortage ratio	0.8084
Shortage denominator	effective_qualified_capacity_kt
Minimum specification-matched shortage	0.1
Sustained period	Not calculable

R7 · Latent domestic capacity

Full · Does not fire

Synthetic utilisation/equivalence does not meet the configured latent-capacity test.

Test no-support, market linkage, procurement, or barrier removal only when equivalence and latent capacity are established.

Effective utilisation	0.89
Maximum effective utilisation	0.7
Specification equivalence	Yes
Qualified available supply	Not calculable

R8 · Committed future demand

Disabled · Not evaluable

Committed and announced layers are disclosed, but base demand, commitment probability, and minimum efficient scale govern R8 calculability.

Do not infer probability-adjusted demand addition or MES fill; keep committed and announced demand separate.

Base demand	Not calculable
Committed demand	74
Announced demand	22
Target-specification demand	104
Downside demand	100
Commitment probability	Not calculable
Probability-adjusted committed demand (kt)	Not calculable
Probability-adjusted demand addition	Not calculable
Minimum probability-adjusted demand addition	0.2
Minimum efficient scale (kt)	50
Minimum efficient scale coverage	Not calculable
Minimum scale-coverage threshold	0.25

Evidence passports and contradictions

4 public records; 14 synthetic records.

Contradiction register

Public evidence contradictions

[S-UNICOIL-SPEC](#): Published coating range differs from EPD and is retained for confirmation.

Synthetic evidence contradictions

No synthetic contradictions are recorded for this simulation.

S-WITS-721049

WITS/UN Comtrade Saudi trade rows for HS 721049

Source field	Value
Source	WITS/UN Comtrade
Class	B
Status	observed
Source period	2021/2023/2024
Retrieval date	2026-08-31
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Frozen annual world rows; concentration and dispersion are disclosed aggregates from the methodology worked case.
Supporting claims	- TRADE_VALUE• TRADE_QUANTITY - SUPPLIER_CONCENTRATION
Contradiction register	Unavailable
Source reference	https://wits.worldbank.org/trade/comtrade/en/country/SAU/year/2024/tradeflow/Imports/partner/ALL/product/721049

S-UNICOIL-EPD

UNICOIL 2024 Environmental Product Declaration

Source field	Value
Source	UNICOIL
Class	C
Status	observed
Source period	2023/2024
Retrieval date	2026-08-31
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Unavailable
Supporting claims	- DOMESTIC_NAMEPLATE_CAPACITY - DOMESTIC_PROCESS_ROUTE - DOMESTIC_SPECIFICATION_ENVELOPE - DOMESTIC_LABORATORY_METROLOGY
Contradiction register	Unavailable
Source reference	https://www.unicoil.com.sa/wp-content/uploads/2026/05/EPD-Report_GS_Unicoil_Rev02-1_1.pdf

S-UNICOIL-SPEC

UNICOIL Arabic and English galvanized product specifications

Source field	Value
Source	UNICOIL
Class	C
Status	observed
Source period	Unavailable
Retrieval date	2026-08-31
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Unavailable

Source field	Value
Supporting claims	- BILINGUAL_SPECIFICATION_EXTRACTION - DOMESTIC_SPECIFICATION_ENVELOPE - DOMESTIC_DIMENSION_ENVELOPE
Contradiction register	Published coating range differs from EPD and is retained for confirmation.
Source reference	https://www.unicoil.com.sa/products%26services/gi-product-brands-specifications-2/

S-HADEED

Hadeed flat-products catalogue

Source field	Value
Source	Hadeed
Class	C
Status	observed
Source period	Unavailable
Retrieval date	2026-08-31
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Unavailable
Supporting claims	DOMESTIC_PROCESS_FAMILY
Contradiction register	Unavailable
Source reference	https://hadeed.com.sa/products?group-category=cold-rolled-galvanized&main-category=flat-products

SYN-MINISTRY-STEEL-001::capability_states

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Capability States

Source field	Value
Source	DEMO_GENERATOR

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: capability_states
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::class_if_confirmed

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Class If Confirmed

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: class_if_confirmed
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::counterfactual

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Counterfactual

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: counterfactual
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::decision_specific_hard_gates

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Decision Specific Hard Gates

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable

Source field	Value
Supporting claims	Simulation branch input: decision_specific_hard_gates
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::demand

SIMULATED — NOT MINISTRY EVIDENCE

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

Demand

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: demand
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::economics

SIMULATED — NOT MINISTRY EVIDENCE

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

Economics

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: economics
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::equivalence

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Equivalence

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: equivalence
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::evsi

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Evsi

Source field	Value
Source	DEMO_GENERATOR

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: evsi
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::hard_exclusion_inputs

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Exclusion Inputs

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: hard_exclusion_inputs
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::hard_gates

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Gates

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: hard_gates
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::plant_line

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Plant Line

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable

Source field	Value
Supporting claims	Simulation branch input: plant_line
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::route_evidence

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Route Evidence

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: route_evidence
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::target_specification

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Target Specification

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: target_specification
Contradiction register	Unavailable

SYN-MINISTRY-STEEL-001::upgrade

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Upgrade

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: upgrade
Contradiction register	Unavailable

Conditions, kill conditions and next facts

Class D simulation · SYN-MINISTRY-STEEL-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Decision conditions

- Achieve customer acceptance for the named target specification.
- Deliver 50 kt of incremental qualified capacity within 18 months.
- Support is milestone-based, sunset-bound and subject to clawback.

Kill conditions

- Committed target-specification demand falls below 72 kt.
- Incumbent output or another expansion closes the gap before award.
- Customer qualification is not achieved by the contractual milestone.

Next evidence actions

- None

Source field	Value	Unit
Approximate EVSI	129.3	SAR m
Route-change probability	0.72	
Value difference (SAR m)	190	
Evidence acquisition cost (SAR m)	2.5	
Delay cost (SAR m)	5	
Approximate EVSI	129.3	
Positive result	Yes	
Next fact to establish	Confirm line-level qualification share and customer acceptance test	

Decision history and expert overrides

None recorded; no completed review is implied.

Authority and reproducibility

Authority and versions

Source reference	Value
Methodology	docs/authority/Industrial_Opportunity_Resolution_Methodology_Final_KSA.docx 5717cbd42acc9947ce5e450013719275acb7ed1470847b21fb2cc547c8ac4ce9
Project	0.2.0
Thresholds	1.3.0
Sector profiles	1.1.0
Evidence policy	1.4.0
UI strings	1.7.0

Evidence boundary

Public decision uses public evidence only	Yes
Synthetic branch isolation	Yes
Scenario reconciliation	
Status	PASS
Scenario identifier	SYN-MINISTRY-STEEL-001
Opportunity identifier	SAU-H0-721049
Public snapshot identifier	PUBLIC-SAU-H0-721049-2026-08-31
Recorded checks	

• Rule identifier	target_spec_demand_within_public_imports
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	target_spec_demand_kt <= latest_public_imports_kt
Recorded result	PASS
Blocking check	Yes

Source inputs	
Target-specification demand	104
Latest public observation year	2024
Latest public imports (kt)	287.9
Source explanation	104 kt <= 287.9 kt.
• Rule identifier	line_nameplate_within_disclosed_public_capacity
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	plant_line.nameplate_kt <= sum(non_null installed_capacity_tpy) / 1000
Recorded result	PASS
Blocking check	Yes
Source inputs	
Line nominal capacity (kt)	250
Disclosed installed capacity (t/year)	◦ 250000◦ Unavailable
Publicly disclosed nominal capacity (kt)	250
Expansion assumption applied	No
Source explanation	250 kt <= 250 kt.
• Rule identifier	capacity_factors_within_unit_interval
Source	Core 06 §10 item 3 lines 226-233; Core 06 §6.2 lines 148-156; Core 05 §8 lines 197-203
Recorded formula	for every capacity factor: 0 <= factor <= 1
Recorded result	PASS
Blocking check	Yes
Source inputs	
Availability factor	0.92
Yield factor	0.94

Qualification share	0.38
Market allocation share	0.7
Current utilisation	0.89
Source explanation	All declared capacity factors are within [0,1].
• Rule identifier	qualified_availability_within_physical_output
Source	Core 06 §5.1 lines 92-101; Core 06 §10 item 3 lines 226-233
Recorded formula	qualified_available_kt <= nameplate_kt * availability * yield
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No qualified_available_kt is declared.
• Rule identifier	demand_layers_remain_separate
Source	Core 04 §2.6 lines 102-112
Recorded formula	observe downside_demand_kt <= target_spec_demand_kt and committed_demand_kt <= target_spec_demand_kt; no decision effect
Recorded result	INFORMATIONAL
Blocking check	No
Source inputs	
Target-specification demand	104
Downside demand	100
Committed demand	74
Base demand	Unavailable
Downside demand within target demand	Yes
Committed demand within target demand	Yes

Source explanation	Demand layers are reported separately; no ordering constraint is enforced.
• Rule identifier	tariff_line_allocation_sums_to_public_hs6_total
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	sum(tariff_line_allocation.lines.quantity_kt) == latest public imports_kt
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No tariff_line_allocation block is declared.
• Rule identifier	buyer_allocation_within_public_imports
Source	Core 06 §5.1 lines 92-101
Recorded formula	sum(buyer_allocation.buyers.quantity_kt) <= latest public imports_kt
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No buyer_allocation block is declared.
• Rule identifier	expansion_assumption_disclosed_and_bounded
Source	Core 06 §5.1 lines 92-101; KL-38
Recorded formula	planned_nameplate_kt >= disclosed public nameplate; commissioning_year >= public as_of year; plant_line.nameplate_kt <= planned_nameplate_kt
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No expansion_assumption block is declared.
• Rule identifier	retained_flows_reconcile_to_public_trade

Source	Core 06 §5.1; methodology §3.3
Recorded formula	retained_imports_kt <= imports_kt; retained + reexports == imports when reexports numeric; domestic_origin_exports_kt <= public exports_kt
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No production_and_retained_flows block is declared.
• Rule identifier	base_demand_and_commitment_probability_valid
Source	Core 06 §5.2; Core 07 §3 R8
Recorded formula	0 <= commitment_probability <= 1; base_demand_kt > 0; minimum_efficient_scale_kt > 0 when declared
Recorded result	PASS
Blocking check	Yes
Source inputs	
Base demand	Unavailable
Commitment probability	Unavailable
Minimum efficient scale (kt)	50
Source explanation	Base demand, commitment probability and MES inputs are valid.
• Rule identifier	shared_enabler_declaration_valid
Source	Core 06 scenario contract 2.1.0 and methodology §8.3
Recorded formula	explicit route valuation and complete Class-D declaration
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No shared-enabler declaration is present.

Latest imports (USD million)	236.9
Latest imports (kt)	287.9
Public decision unchanged after simulation	Yes
Scenario expected-result comparison	
Expected result	
State	ADVANCE
Route code	5
Actual result	
State	ADVANCE
Route code	5
Results match	Yes